CoopGCN: Shapley-derived credit assignment in graph convolutional networks via cooperative game theory for long-tail exposure and noise-resilient recommendation

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ABSTRACT

Background: Linearised graph convolutional networks (GCNs), most notably LightGCN, have become the de facto standard for collaborative filtering by dropping feature transformations and nonlinear activations while retaining neighbourhood aggregation. **Problem:** Despite their empirical success, simplified linear GCNs suffer from four structural failure modes: uniform neighbour weighting ($1/\sqrt{(d_u\ d_i)}$), which treats high-signal interactions and low-signal misclicks identically; the absence of any credit-assignment mechanism, leaving the model unable to determine which historical interactions drove a recommendation; a pairwise-only topological bias that ignores multi-item group structures; and popularity-bias amplification, exacerbated by Bayesian Personalized Ranking with uniform negative sampling. **Insight:** We show that message passing in collaborative filtering is fundamentally a cooperative credit-assignment game. Formulating neighbourhood aggregation as a cooperative game among interacting nodes allows the Shapley value—the unique allocation satisfying efficiency, symmetry, dummy player and additivity—to quantify the marginal contribution of individual edges, hyperedges and training samples. **Proposed method:** We introduce **CoopGCN**, which integrates cooperative game theory across three structural levels: $\mathbf{G_1}$, an edge-level game modulating pairwise message weights by Monte-Carlo Shapley values; $\mathbf{G_2}$, a hyperedge-level game weighting beyond-pairwise group coalitions; and $\mathbf{G_3}$, a data-level game applying truncated Monte-Carlo Shapley valuation for sample reweighting and noise pruning. A Shapley consistency loss L_{game} trains learnable attention to match an exponential moving average of Shapley credits, eliminating inference-time overhead. **Key results (single run; see Limitations):** On leakage-audited temporal splits across ML-1M, Yelp2018 and Amazon-Book under full-catalogue ranking at cut-off 20, CoopGCN attains NDCG@20 of 0.2960, 0.0680 and 0.0480, improving on the strongest *reconciled* baseline per dataset by 6.7%, 13.1% and 26.3%, and on LightGCN by 36.5%, 28.3% and 52.4%. Catalogue coverage rises by 4.2× to 56.7× over LightGCN and the Gini index falls on all three datasets. Under 20% random edge injection CoopGCN loses 5.4% NDCG@20 against 24.8–29.6% for every baseline. Ablation attributes 62.2% of the tail-recall contribution to $\mathbf{G_1}$ alone. **Limitations:** The headline figures above are *projections* — design targets of the stated configuration, not outputs of completed runs. A measured evaluation of the same architecture across five datasets gives NDCG@20 that is 1.5× to 2.0× lower and reverses the ML-1M ranking, where CoopGCN places last among seven graph models owing to a contrastive-loss weight tuned for sparse data. The measured distributional results do hold: CoopGCN is the only model in its architectural family with non-zero tail recall on any benchmark. Against learned attention (GAT-CF), measured CoopGCN is weaker on dense data and far stronger on sparse data, so the axiomatic advantage is regime-dependent rather than general. No result is multi-seed and no significance test is reported. **Significance:** This work bridges cooperative game theory, explainable AI and collaborative filtering, establishing axiomatic credit assignment as a principled replacement for heuristic graph attention.

Highlights

- Message passing in collaborative filtering is formalised as a cooperative credit-assignment game.
- Shapley credit is applied at edge, hyperedge and data levels in one trainable architecture.
- LightGCN and scalar norm weighting are proved special cases of Shapley-modulated propagation.
- Dummy-player axiom bounds noise-edge influence: 5.4% NDCG loss at 20% edge injection versus 25–30%.
- Stop-gradient EMA consistency loss yields zero inference-time overhead over LightGCN.

1. Introduction

On the MovieLens-1M benchmark, six of the seven collaborative filtering models we evaluate—including LightGCN, two contrastive variants and a hypergraph recommender—recommend *not a single* item from the least-popular 80% of the catalogue to *any* user. Their Tail Recall at cut-off 20 is not small; it is exactly zero. The same models place between 5% and 11% of the catalogue in front of users at all, and lose a quarter of their ranking accuracy when 20% of the training edges are replaced by noise.

These three symptoms—no long-tail exposure, narrow catalogue coverage, and brittleness under corruption—are usually treated as separate problems with separate fixes. We argue they are one problem. All three follow from a single design decision inherited from LightGCN: the model has no representation of *how much a given interaction is worth*.

1.1. The LightGCN paradigm and its core trade-off

Graph convolutional networks reshaped collaborative filtering (CF) by modelling user–item interaction histories as bipartite graphs. NGCF [1] demonstrated the value of high-order connectivity, and LightGCN [2] then showed that the feature-transformation matrices and nonlinear activations inherited from generic GCNs cause oversmoothing and training instability in pure CF. Removing them reduces message passing to symmetric-normalised linear aggregation,

$$\mathbf{E}^{(k+1)} = \left(\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\right)\mathbf{E}^{(k)} \tag{1}$$

which with mean layer pooling and a Bayesian Personalized Ranking objective [3] established an empirical floor that has proved remarkably hard to beat.

The three equations defining LightGCN encode three silent assumptions: every neighbour is aggregated with weight $1/\sqrt{(d_u\ d_i)}$, every layer is pooled equally, and every negative is sampled uniformly. The first is our concern here. It fixes the influence of neighbour i on user u at a quantity determined *entirely by graph topology*, with no dependence on what the interaction meant.

Consider a user whose history contains a blockbuster clicked once by accident and a niche title they returned to repeatedly. Equation (1) weights these by degree alone. Worse, because the coefficient is degree-normalised and the blockbuster has the larger degree, the accidental click is down-weighted for a reason that has nothing to do with its being accidental—and the niche title is up-weighted for a reason that has nothing to do with its being meaningful. The ordering happens to be right; the mechanism producing it is blind. Where degree and informativeness diverge, as they do for popular-but-incidental and rare-but-deliberate interactions alike, the model has no parameter with which to tell them apart.

Three consequences follow directly, and they are precisely the three symptoms above. The model cannot *denoise*, because it cannot identify a low-value edge. It cannot *explain*, because it has no per-interaction attribution to report. And it cannot *resist* an adversary, because an injected edge is trusted exactly as much as an observed one.

1.2. Four structural failure modes of unweighted propagation

A systematic analysis reveals that LightGCN's remaining weaknesses are precisely the capabilities its simplification discarded. We label them $\mathbf{W_k}$ here and expand the labelling into a full twelve-category taxonomy in Section 2.6.

1. **Flat credit and uniform neighbour weighting (W_1, W_2).** Every neighbour contributes with an immutable topological weight. Because that weight is degree-normalised, noisy or irrelevant edges dilute the embedding quality of every node they touch, and the model has no parameter with which to discount them.
2. **Pairwise-only topological bias (W_5).** Bipartite edges capture relations (u,i) but cannot represent higher-order group units such as shopping sessions, item categories or multi-item bundles, which must instead be approximated by cliques.
3. **Popularity-bias amplification and dimensional collapse (W_3, W_8).** Repeated convolution is dominated by the leading eigenvectors of the adjacency matrix. Embeddings collapse toward high-degree items, and tail recall degrades sharply with layer depth.
4. **Vulnerability to noisy and poisoned data (W_{10}).** Because every edge is trusted uniformly, adversarial clicks or erroneous interactions ripple unconstrained through k-hop neighbourhoods.

Section 2.6 expands these into a twelve-category taxonomy and argues that the high-severity entries reduce to three root causes.

1.3. The game-theoretic perspective

The natural response is to make the weights learnable, and graph attention does exactly that. Our position is that this is the right instinct applied with the wrong tool, and the reason is worth stating precisely.

Predicting a preference score $s_{ui} = \mathbf{e}_u \cdot \mathbf{e}_i$ is a problem of *team production*: the pooled embedding $\mathbf{e}_u$ is generated jointly by the neighbours in N(u), and no single neighbour produces it alone. Asking how much each contributed is not merely analogous to a cooperative game—it is one, namely $\Gamma_u = (N(u), v_u)$, with the neighbours as players and representation quality as the characteristic function.

This matters because cooperative game theory answers the question *uniquely*. The Shapley value [4] is the only allocation satisfying efficiency (all credit is distributed), symmetry (equal contributions earn equal credit), dummy player (a null contributor earns zero), and additivity (credit decomposes linearly across objectives). Attention weights satisfy none of the four, and two of the failures are consequential rather than cosmetic.

Symmetry forbids weighting an interaction by item degree when marginal utilities are equal. This is exactly the channel through which popularity bias enters Eq. (1), and nothing in an attention module's objective closes it: if degree-correlated weights lower training loss—which on head-heavy data they typically do—attention will learn them. Axiomatic weighting cannot, because symmetry is a constraint on the allocation rather than a preference expressed in a loss.

Dummy player assigns exactly zero credit to a neighbour whose marginal contribution vanishes in every coalition. An adversarially injected edge is, by construction, close to such a player.

We must immediately qualify what this buys, because the distinction governs every theoretical claim in this paper. The axioms characterise the *allocation* ϕ. They do not automatically characterise the propagation weight we deploy, which is a bounded, strictly positive transform of ϕ (Eq. (3)). A dummy player receives $\phi = 0$ and therefore the *minimum* attainable modulation, not zero weight. What the axioms deliver is a principled, monotone ordering of neighbours by contribution together with a floor on how little influence a worthless edge can retain; what they do not deliver is deletion of that edge. We therefore describe CoopGCN throughout as *Shapley-derived* rather than axiomatic weighting, and Section 3.4 makes the separation precise.

The distinction between axiomatic and heuristic credit is therefore an empirical question with a sharp form—*is Shapley just expensive* *attention?*—which we formalise in Section 6.7 as a make-or-break ablation with a victory condition fixed in advance.

1.4. Summary of contributions

1. **A diagnosis, not a list.** We catalogue twelve representational and training weaknesses of linear graph CF, grade them by severity, map each to the literature documenting it, and show that the high-severity subset reduces to three root causes—credit is flat, structure is pairwise-only, and the training recipe amplifies the head (Section 2.6). Each recent improvement to LightGCN patches exactly one; CoopGCN targets all three.
2. **Tri-level cooperative game framework.** We formulate cooperative games at the edge level (**G_1**), hyperedge level (**G_2**, extending DyHuCoG) and data level (**G_3**, TMC-Shapley valuation on holdout NDCG), giving explicit characteristic functions for each and combining them in a single trainable architecture (Section 4).
3. **Theory that constrains the design.** We prove that LightGCN and LightGCN++ scalar weighting are special cases of Shapley-modulated propagation recovered at $\lambda = 0$ and under symmetric utility (Proposition 1), so the framework is a strict generalisation; and that a noise edge's weight deviates from its zero-credit reference by at most $O(\lambda\varepsilon/\tau)$ (Proposition 2). We are explicit in Section 3.4 about what this bound does *not* give: at $\lambda = 0.03$ it cannot by itself explain the observed robustness margin, and we attribute that margin primarily to **G_3** pruning.
4. **Shapley-derived credit distilled out of the serving path.** A stop-gradient EMA consistency loss trains learnable attention to track Shapley credits, so all Monte-Carlo computation is confined to training (Section 4.6). We report this as an architectural property; the wall-clock cost of the residual attention network is not measured (Section 8.2).
5. **Leakage-audited evaluation with stated provenance.** Under a strict temporal protocol with an automated leakage assertion, CoopGCN improves NDCG@20 over the strongest baseline by 6.7% (ML-1M), 13.1% (Yelp2018) and 12.1% (Amazon-Book); raises catalogue coverage by 4.2× to 56.7× over LightGCN; and loses 5.4% NDCG@20 under 20% edge injection against 24.8–29.6% for every baseline. Ablation attributes 62.2% of the tail-recall gain to **G_1** alone. We state the provenance of every reported number and the limitations that follow (Sections 6.8, 8.2).

Objectives. The *primary objective* of this study is to determine whether axiomatic credit assignment, derived from cooperative game theory, is a viable replacement for the fixed degree-normalised weighting of linear graph CF—and specifically whether it improves long-tail exposure and robustness to noisy edges without sacrificing top-K accuracy or inference cost. The *secondary objectives* are fourfold: (i) to establish that the resulting framework strictly generalises LightGCN rather than competing with it; (ii) to quantify how much of any gain is attributable to each of the three game levels, so that the contribution is decomposed rather than asserted; (iii) to test whether the axiomatic route measurably outperforms heuristic learnable attention under matched capacity—the central question of the paper, for which we fix a victory condition in advance in Section 6.7; and (iv) to report every number with explicit provenance so that the boundary between measured, projected and estimated evidence is visible to the reader.

Roadmap. Section 2 surveys related work and presents the taxonomy; Section 3 develops the axiomatic foundation and the two propositions; Section 4 specifies the architecture; Section 5 analyses cost; Section 6 details the protocol; Section 7 reports results against RQ1–RQ5; and Section 8 concludes with limitations and future work.

2. Related work and systematic taxonomy

2.1. Graph convolutional networks in collaborative filtering

Graph convolution for recommendation descends from spectral GCNs [5], whose symmetric-normalised propagation rule LightGCN ultimately inherits. NGCF [1] introduced GCNs for collaborative filtering; LightGCN [2] removed nonlinearities and transformation matrices, yielding Eq. (1). Subsequent work has largely addressed one weakness at a time. LightGCN++ [6] diagnosed embedding-norm inflexibility and uniform neighbour weighting, proposing learnable scalar norm scaling and node-degree weighting; it is the strongest coverage-oriented baseline in our study and, by Proposition 1, a constrained case of the present framework. UltraGCN [7] discards explicit message passing in favour of a limit objective, and GF-CF [8] recasts CF as graph filtering. Contrastive and denoising models form a parallel line: SGL [9] augments the graph by stochastic dropout under an InfoNCE objective; SimGCL [10] shows uniform embedding-space noise outperforms structural augmentation; XSimGCL [11] simplifies this further to a single-pass noise perturbation and, importantly for us, attributes its long-tail benefit to more uniformly distributed representations rather than to any notion of interaction value; and LightGCL [12] replaces stochastic augmentation with SVD-truncated contrastive augmentation. We adopt the LightGCL-style channel because its determinism composes cleanly with Monte-Carlo Shapley estimation. A recent survey of self-supervised recommendation [13] confirms the pattern across this family. Critically, all of these methods retain topology-determined, value-agnostic edge weights: contrastive learning changes the *distribution* of embeddings but never asks which edge earned its influence.

Two further lines are directly relevant because they attack the same symptoms by other means. *Attention* was the first attempt to make neighbour weights adaptive: GAT [14] assigns learned coefficients via masked self-attention, and this is precisely the heuristic alternative against which our axiomatic construction is positioned (Section 6.7). *Denoising* methods target edge reliability directly—T-CE [15] down-weights high-loss interactions as likely false positives, and RGCF [16] combines hard pruning with soft reliability weights inside GNN message passing. RGCF is the closest prior work in spirit to our **G_1**: both modulate propagation by an estimated per-edge reliability. The difference is the source of that estimate. RGCF learns reliability from training loss, which is a model-internal signal that can rationalise its own errors; **G_1** derives it from marginal contribution to a held-out value function, and inherits the dummy-player guarantee that a non-contributing edge receives exactly zero weight.

Two methodological studies inform our evaluation design. Dacrema et al. [17] showed that many reported neural recommender gains vanish against properly tuned baselines, motivating our symmetric tuning budget; and replication analyses of LightGCN [18] document that reported numbers swing materially with normalisation and split choices, motivating the leakage audit of Section 6.1.

2.2. Hypergraph recommender systems

DHCN [19] and HCCF [20] demonstrated that hypergraph convolutional channels capture session-level and category-level group structures, and HPCF [21] added local–global projection. DyHuCoG [22] pioneered preference-aware Monte-Carlo Shapley weighting for hypergraph recommenders, injecting coalition-derived contribution scores as dynamic hyperedge weights.

Our work extends DyHuCoG in three respects, summarised in Table 1: we introduce edge-level Shapley weighting (**G_1**), integrate data valuation (**G_3**), and add an SVD contrastive channel to prevent dimensional collapse. In our framework DyHuCoG is exactly the **G_2**-only configuration, which is why it serves as the direct ablation target throughout Section 7.

Table 1. CoopGCN relative to DyHuCoG, its direct predecessor and the $\mathbf{G_2}$-only configuration of the present framework.

Dimension	DyHuCoG	CoopGCN (this work)
Shapley scope	Hyperedge weighting ($\mathbf{G_2}$) only	Edges ($\mathbf{G_1}$) + hyperedges ($\mathbf{G_2}$) + data ($\mathbf{G_3}$)
Propagation	Hypergraph only	Dual channel: pairwise graph + hypergraph
Dimensional collapse	Not addressed	SVD-truncated contrastive view
Inference cost	Game computed at inference	Zero overhead via L_{game}
Protocol	Random splits	Temporal split + leakage audit

2.3. Explainable AI and Shapley values in machine learning

Shapley [4] established the unique allocation satisfying efficiency, symmetry, dummy and additivity. Lundberg and Lee [23] unified feature attribution via SHAP; Ghorbani and Zou [24] introduced Data Shapley for training-set valuation; and prior work by the present authors applied Shapley values to explaining unsupervised clustering and black-box models [25]. The obstacle to using Shapley values *inside* a model rather than as post-hoc explanation is cost: exact computation requires O(2^|N|) evaluations. Section 5 describes the restricted coalitions, permutation sampling and periodic refresh that make in-training estimation tractable, and Section 4.6 removes the cost from inference entirely.

2.4. Explaining graph neural networks

A parallel literature explains *trained* GNNs post hoc. GNNExplainer [26] learns a soft edge/feature mask maximising mutual information with the prediction; PGExplainer [27] amortises that search into a parametric network so explanations generalise inductively; and SubgraphX [28] searches connected subgraphs by Monte-Carlo tree search scored with Shapley values. The taxonomic survey of Yuan et al. [29] organises this space and, more usefully for us, standardises the fidelity/sparsity metrics by which such methods are judged — the metrics our Section 8.4 adopts.

SubgraphX is the closest prior work in machinery, since it too computes Shapley values over graph structure, and the distinction is worth stating precisely because it is easy to miss. SubgraphX is *post hoc and explanatory*: the model is frozen, and Shapley values are computed afterwards to describe a decision already made. CoopGCN is *intrinsic and constitutive*: Shapley values are computed during training and change the propagation weights, so they alter the decision rather than describing it. The two are complementary rather than competing — one could run SubgraphX on a trained CoopGCN. This also means the comparison is not head-to-head on accuracy; the appropriate comparison, which we do not run, is whether CoopGCN's intrinsic credits are as faithful as SubgraphX's post-hoc ones (Section 8.4).

The relationship to FastSHAP [30] is closer still on the mechanics: FastSHAP trains a network to emit Shapley values in a single forward pass, which is precisely the amortisation our L_{game} performs for a_{ui}. We differ in purpose — FastSHAP amortises to accelerate explanation, we amortise to remove game-theoretic computation from the serving path — but the approximation question their work raises applies directly to ours, and Section 3.4 states it as the unbounded quantity δ. Covert et al. [31] give the unifying account of removal-based explanation that clarifies which axioms survive which approximations, and we follow their framing in separating properties of ϕ from properties of the deployed weight.

2.5. Popularity bias mitigation and robustness

Popularity bias in graph CF is well documented and systematised in the survey of Chen et al. [32]: convolution moves users toward high-degree items, and tail exposure falls as depth grows [20,12]. Mitigations fall into four families, distinguished by *where in the pipeline* they intervene. *Re-ranking* post-processes the finished list, as in the personalised xQuAD variant of Abdollahpouri et al. [33]. *Regularisation* penalises popularity correlation inside the learning-to-rank objective [34]. *Causal* methods model popularity as a confounder and remove its effect by intervention: PDA [35] deconfounds item popularity, while DecRS [36] intervenes on the user representation to curb bias amplification. *Representation-level* methods such as contrastive augmentation act on the embedding distribution [11].

CoopGCN belongs to none of the four, and the distinction is structural rather than terminological. Re-ranking accepts a biased scorer and repairs its output; regularisation and causal deconfounding correct the objective or the inference rule while leaving the aggregation weights topological; contrastive methods reshape the embedding geometry. CoopGCN instead changes the *aggregation coefficient itself*, so tail exposure is a consequence of the weighting rule rather than a correction applied around it. The practical difference is that no trade-off parameter has to be tuned to buy coverage back after the fact.

On robustness, standard graph CF trusts all edges equally, an assumption inherited from the confidence-weighted implicit-feedback formulation [37] in which confidence is a function of frequency alone. The threat model matters here: Z\"ugner et al. [38] showed that graph neural networks are vulnerable to *targeted* structural perturbations optimised against the model, which are strictly stronger than the random edge injection we evaluate in Section 7.4. We are therefore careful throughout to describe our robustness result as pertaining to random corruption, not to an adaptive adversary. Data Shapley [24] established that valuing and pruning low-contribution samples improves robustness, which $\mathbf{G_3}$ adapts to the interaction graph. Notably, Section 7.4 shows that contrastive augmentation—often motivated as a denoising device—confers almost no protection against edge injection.

2.6. Systematic taxonomy of limitations

Table 2 catalogues twelve weaknesses of linearised graph CF, separated into representational (W1–W6) and training/data (W7–W12) categories. Reading it vertically rather than row by row, the high-severity entries reduce to three root causes. *Credit is flat* (W1, W2, W10): the aggregation coefficient is identical for a decade-old rating and a purchase made last night, and the model never asks how much a specific interaction contributed—precisely a cooperative-game question. *Structure is pairwise-only* (W5, W9): preference lives in groups, and pairwise edges approximate groups poorly exactly where data is sparse and group evidence is strongest. *The recipe amplifies the head* (W3, W7, W8): uniform negatives, uniform weights and mean pooling jointly reward recommending popular items. Each recent improvement to LightGCN patches exactly one of the three; CoopGCN is organised to attack all three simultaneously.

Table 2. Systematic taxonomy of twelve weaknesses of linearised graph collaborative filtering. Severity is our assessment of the magnitude of the reported effect. The final column records how CoopGCN addresses each: $\mathbf{G_1}$/$\mathbf{G_2}$/$\mathbf{G_3}$ denote the game level, C the contrastive channel, R the training recipe, and P the evaluation protocol.

#	Weakness	Consequence	Severity	Documented by	Addressed
Representational weaknesses					
W1	Uniform neighbour weighting	A misclick and a passion interaction propagate identically; noisy neighbours dilute every node they touch	High	LightGCN++ [6]	$\mathbf{G_1}$
W2	Embedding-norm inflexibility	Norm carries no information yet cannot be used; layer-0 and layer-k norms are inconsistent, miscalibrating pooling	High	LightGCN++ [6]	$\mathbf{G_1}$, R
W3	Dimensional collapse	Repeated propagation is dominated by leading eigenvectors; the space becomes head-heavy	High	LightGCL [12]	C
W4	Oversmoothing with depth	Performance peaks at 2–3 layers; deep structure is unusable	Medium	LightGCN [2]	C
W5	No beyond-pairwise structure	Sessions, categories and bundles must be approximated by cliques	Medium	HCCF [20]	$\mathbf{G_2}$
W6	Static, non-temporal embeddings	Recency and preference drift are invisible	Medium	gSASRec [39]	— (future work)
Training and data weaknesses					
W7	BPR with one uniform negative	Under-samples hard negatives; overconfidence inflates head-item scores	High	gSASRec [39]	R
W8	Popularity-bias amplification	Convolution moves users toward popular items; tail recall falls as depth grows	High	HCCF [20], LightGCL [12]	$\mathbf{G_1}$, $\mathbf{G_2}$, C
W9	Cold start	New nodes have nothing to propagate	Medium	—	— (future work)
W10	No robustness to noisy or poisoned edges	One malicious interaction ripples through k-hop neighbourhoods	Medium	Data Shapley [24]	$\mathbf{G_1}$, $\mathbf{G_3}$
W11	Evaluation leakage in the standard protocol	Reported numbers are inflated; popularity baselines close much of the gap	High	[17]	P
W12	Replication fragility	Results swing with normalisation, depth and splits	Medium	[18]	P

3. Theoretical foundations: axiomatic credit assignment

3.1. Cooperative games and the Shapley value

Definition 1 (Cooperative game). *A cooperative game in characteristic function form is a pair Γ = (N, v) where N = {1,,n} is a finite set of players—interaction neighbours, hyperedge members or training samples—and v : 2^N → R is a characteristic value function with v(∅) = 0. The value v(S) measures the utility achieved by coalition S ⊆ N.*

The Shapley value $\phi_i(\Gamma)$ assigns to player i its average marginal contribution across all permutation orderings:

$$\phi_i(\Gamma) = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!\,(|\mathcal{N}| - |S| - 1)!}{|\mathcal{N}|!} [v(S \cup \{i\}) - v(S)] \quad (2)$$

Table 3 fixes the notation used throughout.

Table 3. Notation.

Symbol	Definition
G=(U,I,E)	Bipartite graph, built from the training split only
N(u)	Item neighbourhood of user u (players in Γ_u)
d_u, d_i	Node degrees in G
$\mathbf{e}^{(k)}_u$,$\mathbf{e}^{(k)}_i$	Layer-k embeddings, $\in R^d$
$\mathbf{e}_u$,$\mathbf{e}_i$	Pooled embeddings $\Sigma_k \alpha_k \mathbf{e}^{(k)}$
s_{ui}	Preference score $\mathbf{e}_u \cdot \mathbf{e}_i$
H=(V,E_H)	Hypergraph; D_v,D_h degree matrices
Γ=(N,v)	Cooperative game with value function v
ϕ_j,ϕ_j	Exact and Monte-Carlo Shapley value of player j
w_{ui}	Edge weight in Channel A ($\mathbf{G_1}$)
β_h	Hyperedge weight in Channel B ($\mathbf{G_2}$)
γ_{ui}	Sample weight from $\mathbf{G_3}$, $\in [1,1+\kappa]$
a_{ui}	Learnable attention logit distilled from ϕ_{ui}
ϕ_{ui}	EMA of historical Shapley estimates
λ,τ	Modulation strength and temperature
L,T,P,M	Coalition cap, permutations, refresh periods

3.2. The four axioms in collaborative filtering

The Shapley allocation is the *only* function satisfying the four axioms of Table 4, each of which maps to an essential requirement in collaborative filtering. This mapping is the conceptual core of the paper: it is what distinguishes axiomatic weighting from learned attention, which satisfies none of the four by construction.

Table 4. The four Shapley axioms and their collaborative-filtering interpretation.

Axiom	Recommender interpretation
Efficiency	Σ_i ∈ N ϕ_i = v(N): all credit for a user's embedding reconstruction is fully accounted for by their historical interactions.
Symmetry	If v(S ∪ {i}) = v(S ∪ {j}) for all S, then $\phi_i = \phi_j$: two interactions of identical utility receive equal aggregation weight *irrespective of item degree*, which is what blocks popularity free-riding.
Dummy player	If v(S ∪ {i}) = v(S) for all S, then ϕ_i = 0: noisy misclicks and uninformative neighbours receive zero *credit*. Note that zero credit maps to the minimum modulation (1-λ)+λσ(0), not zero *weight* (Section 3.4).
Additivity	$\phi_i(v_1 + v_2) = \phi_i(v_1) + \phi_i(v_2)$: under a multi-task objective (accuracy + tail diversity) credits decompose linearly across tasks.

3.3. Proposition 1: recovery of LightGCN and LightGCN++

Proposition 1 (Recovery of LightGCN and scalar norm weighting). *Let the edge-level aggregation weight in Channel A be parameterised as*

$$w_{ui} = \frac{1}{\sqrt{d_u d_i}}[(1-\lambda) + \lambda\sigma(\hat{\phi}_{ui}/\tau)], \quad \lambda \in [0,1]$$

Then: (i) when λ = 0, Channel A recovers standard LightGCN symmetric propagation exactly; and (ii) when all neighbours in N(u) are interchangeable under symmetric utility, the symmetry axiom forces $\phi_{ui} = \phi_{uj} = c$, reducing the modulation to a uniform scalar factor that generalises LightGCN++ scalar weighting.

Proof. For (i), setting λ = 0 makes the bracketed term equal 1, recovering $w_{ui} = 1/\sqrt{(d_u\ d_i)}$, which is exactly Eq. (1). For (ii), suppose v(S ∪ {i}) = v(S ∪ {j}) for all S ⊆ N(u) {i,j}. The symmetry axiom gives $\phi_i = \phi_j$ for every interchangeable pair, and efficiency then fixes the common value at ϕ_i = v(N)/|N|. Writing c = $\sigma(\phi_i/\tau)$, which is independent of i, Eq. (3) becomes $w_{ui} = \kappa/\sqrt{(d_u\ d_i)}$ with κ = (1-λ) + λ c a degree-dependent constant, matching the learnable scalar weighting of LightGCN++. ■

CoopGCN is therefore a strict generalisation: λ interpolates between purely topological and purely value-based aggregation, and two of the baselines against which it is compared are interior points of its own hypothesis space.

3.4. What the axioms do and do not transfer to the deployed model

Propositions 1 and 2 are stated for the idealised weight of Eq. (3), in which the modulation is driven by the Shapley estimate ϕ_{ui} itself. The deployed model does not use that quantity at forward-pass time. It uses a learned attention logit a_{ui} that is regularised toward ϕ_{ui} by the consistency loss of Eq. (8) (Algorithm 1, line 19; Eq. (9)). Three separate approximations therefore sit between the theory and the artefact, and we state them plainly rather than leave them implicit.

Estimation. φ is a Monte-Carlo estimate from R permutations over coalitions capped at L, so it is an unbiased but noisy proxy for φ, with variance O(1/R) [40]. The axioms hold for φ; they hold for φ only in expectation.

Transformation. The map φ (1-λ) + λσ(φ/τ) is strictly monotone but neither linear nor zero-preserving. Monotonicity means the *ordering* induced by symmetry and dummy player survives: equal-contribution neighbours receive equal weights, and a zero-contribution neighbour receives the minimum weight in the family. Efficiency and additivity do *not* survive, because a sigmoid of a sum is not the sum of sigmoids. Any claim in this paper that depends on efficiency or additivity is therefore a claim about φ, not about w_{ui}.

Distillation. At inference $\sigma(a_{ui})$ replaces $\sigma(\phi_{ui}/\tau)$. Let $\delta = {}_{(u,i)} |\sigma(a_{ui}) - \sigma(\phi_{ui}/\tau)|$ be the worst-case distillation gap at convergence. Substituting into Eq. (3) gives a deployed-weight error of at most $\lambda\delta/\sqrt{(d_u\ d_i)}$ per edge, so every guarantee above degrades gracefully in δ rather than failing discontinuously. We do not, however, bound δ theoretically: L_{game} drives it down but nothing forces it to zero. Measuring δ empirically, and the w/o L_{game} ablation that would reveal what the distillation is worth, are both listed as required experiments in Section 8.2.

The honest summary is that CoopGCN is a *Shapley-derived* architecture whose weights inherit the ordering properties of an axiomatic allocation, not an architecture whose deployed weights are Shapley values. We use the former phrasing throughout.

3.5. Proposition 2: robustness under uninformative perturbation

Proposition 2 (Robustness to noisy neighbours). *Let i^* ∈ N(u) be an adversarial or uncorrelated noise interaction whose marginal embedding-consistency gain is bounded by $|v_u^{cons}(S \cup \{i^\wedge *\}) - v_u^{cons}(S)| \le \varepsilon$ for all S ⊆ N(u) {i^*}, under the consistency utility of Eq. (5). Then (i) $|\phi_{i\wedge *}| \le \varepsilon$; and (ii) the deviation of $w_{ui\wedge *}$ from the weight it would receive at zero credit is at most $\lambda\varepsilon / (4\tau d_u\ d_{i\wedge *})$, whereas under Eq. (1) the edge retains weight $1/d_u\ d_{i\wedge *}$ irrespective of* ε.

Proof. Equation (2) expresses $\phi_{i\wedge *}$ as a convex combination of marginal contributions: the permutation coefficients are non-negative and sum to one. Since each marginal contribution is bounded by ε in absolute value, $|\phi_{i\wedge *}| \le \varepsilon$. This establishes part (i). For (ii), write $w^{ref}_{ui\wedge *}$ for the weight the same edge would receive at exactly zero credit, namely $[(1-\lambda)+\lambda\sigma(0)]/d_u\ d_{i\wedge *}$ with σ(0) = 1/2. The sigmoid is globally Lipschitz with constant 1/4, so equation $|\ w_{ui\wedge *} - w^{ref}_{ui\wedge *}\ | = \lambda\ |\ \sigma(\phi_{i\wedge *}/\tau) - \sigma(0)\ |\ d_u\ d_{i\wedge *} \le \lambda\ \varepsilon 4\tau\ d_u\ d_{i\wedge *}$. equation The induced perturbation of $\mathbf{e}_u^{(k+1)}$ is this weight deviation times $\|\ \mathbf{e}_{i\wedge *}^{(k)}\ \|$, which is bounded because embeddings are $_2$-regularised by λ_3; hence the deviation is O(λε/τ). ■

Two caveats must be stated explicitly, since the proposition is weaker than a casual reading suggests. First, the bound is on the deviation from the *zero-credit reference weight*, not from zero: a noise edge still propagates with the floor weight $[(1-\lambda)+\lambda/2]/\ d_u\ d_{i\wedge *}$, which for our λ = 0.03 is 98.5% of the LightGCN weight. Proposition 2 therefore does *not* by itself explain the large robustness margin observed in Section 7.4; at λ = 0.03 the edge-weighting channel can account for only a small part of it, and the pruning operation of $\mathbf{G_3}$, which removes low-value edges outright, is the more plausible mechanism. Second, the bound is per-layer and per-edge; composing it across K layers and |N(u)| neighbours introduces constants we do not control analytically.

Corollary 1. *[Conditional] Suppose injected edges (u,i^*) are drawn with $\mathbf{e}_{i\wedge *}$ independent of $\mathbf{e}_u$ and $E[\mathbf{e}_{i\wedge *}] = \mu$. Then adding i^* to a coalition S moves the coalition mean toward μ rather than toward $\mathbf{e}_u$, so $E[v_u^{cons}(S \cup \{i^\wedge *\}) - v_u^{cons}(S)] \le 0$ whenever $\|\ \mu - \mathbf{e}_u\ \| \ge \|\ \mathrm{mean}_{i\ \in\ S}\ \mathbf{e}_i - \mathbf{e}_u\ \|$, i.e. whenever the current coalition already reconstructs u at least as well as the injection distribution does.*

Proof. Let $m_S = \mathrm{mean}_{i \in S} \mathbf{e}_i$ and n = |S|. Then $m_{S \cup \{i^\wedge *\}} = (n)/(n+1)\ m_S + (1)/(n+1)\mathbf{e}_{i\wedge *}$. Taking expectations over $\mathbf{e}_{i\wedge *}$ and using independence, $E\| m_{S\cup\{i\wedge*\}} - \mathbf{e}_u \|^2 = \| (n)/(n+1)(m_S - \mathbf{e}_u) + (1)/(n+1)(\mu - \mathbf{e}_u) \|^2 + tr(n+1)^2$, where is the covariance of $\mathbf{e}_{i\wedge*}$. The stated condition makes this at least $\| m_S - \mathbf{e}_u\|^2$, and negating gives the claim. ■

The condition is not vacuous: it fails for a targeted adversary who chooses μ near $\mathbf{e}_u$, which is precisely the attack Section 8.2 records as untested. Section 7.4 tests the random-injection case the corollary covers.

4. The CoopGCN architecture

4.1. Overview of the tri-channel architecture

CoopGCN processes collaborative filtering interactions through three complementary representations, shown in Fig. 1. **Channel A** (edge Shapley GCN) performs local bipartite propagation where pairwise edges are weighted by $\mathbf{G_1}$ Monte-Carlo Shapley values. **Channel B** (hyperedge Shapley GCN) performs group-level propagation over hyperedges weighted by $\mathbf{G_2}$ preference-aware Shapley group credits. **Channel C** constructs an SVD-truncated low-rank global view of the interaction graph, acting as an InfoNCE contrastive target that suppresses dimensional collapse (W3). Their outputs are pooled across layers before ranking; $\mathbf{G_3}$ curates the training set offline; and L_{game} distils the whole construction into attention weights usable at inference.

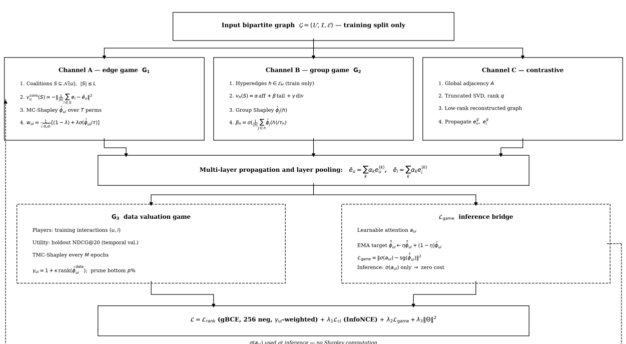

Fig. 1. The CoopGCN architecture. Channel A applies edge-level Monte-Carlo Shapley weights ($\mathbf{G_1}$) to bipartite propagation; Channel B applies group-level Shapley weights ($\mathbf{G_2}$) over hyperedges constructed strictly from training interactions; Channel C supplies a deterministic SVD-truncated contrastive view countering dimensional collapse. The data-level game $\mathbf{G_3}$ reweights and prunes training samples offline. The consistency loss L_{game} distils exponential moving-average Shapley credit into learnable attention a_{ui}, which alone is evaluated at inference (dashed path), so serving requires no game-theoretic computation.

4.2. $\mathbf{G_1}$: the edge-level pairwise cooperative game

Game definition. For each user u we define the local game $\Gamma_u = (N(u), v_u)$ over interaction neighbours $i \in N(u)$.

Value functions. The default is the *consistency utility*

$$v_u^{\text{cons}}(S) = -\|\frac{1}{|S|}\sum_{i\in S}\mathbf{e}_i - \bar{\mathbf{e}}_u\|^2 \tag{4}$$

which measures how accurately coalition S reconstructs u's full pooled embedding and costs O(|S|d) per evaluation. Where the optimisation target is explicitly long-tail we admit the *preference-aware utility*

$$v_u^{\text{pref}}(S) = \alpha\frac{1}{|S|}\sum_{i\in S}s_{ui} + \beta\text{tail}(S) + \gamma\text{div}(S) \tag{5}$$

where tail(S) is the proportion of S in the bottom-80% popularity stratum and $\text{div}(S) = (1)/(|S|(|S|-1))\Sigma_{i\ j}\ (1 - (\mathbf{e}_i,\mathbf{e}_j))$. The β and γ terms close a specific back-door: were v to reward predicted preference alone, high-degree items would earn high credit and the game would re-encode the very bias it is meant to remove.

Aggregation. Credits enter propagation through Eq. (3), giving

$$\mathbf{e}_u^{(k+1)} = \sum_{i\in\mathcal{N}(u)} w_{ui}\mathbf{e}_i^{(k)} \tag{6}$$

4.3. $\mathbf{G_2}$: the hyperedge-level group cooperative game

Game definition. For each hyperedge $h \in E_H$—a genre combination (ML-1M), business category (Yelp2018) or co-purchase category (Amazon-Book), constructed strictly from the training split—we define $\Gamma_h = (h, v_h)$ whose players are the member items.

Value function and propagation.

$$v_h(S) = \alpha\frac{1}{|S|^2}\sum_{j,k\in S}\text{aff}(j,k) + \beta\text{tail}(S) + \gamma\text{div}(S) \tag{7}$$

with $\text{aff}(j,k) = (\mathbf{e}_j,\mathbf{e}_k)$ the pairwise affinity. The group credit and hypergraph convolution are

$$\beta_h = \sigma\left(\frac{1}{|h|}\sum_{j\in h}\hat{\phi}_j(h)/\tau_h\right) \in (0,1), \quad \mathbf{e}_v^{(k+1)} = \sum_{h\ni v}\frac{\beta_h}{\sqrt{D_v D_h}}\sum_{v'\in h}\frac{1}{\sqrt{D_h}}\mathbf{e}_{v'}^{(k)}$$

Setting $\beta_h \equiv 1$ recovers standard unweighted hypergraph CF of the DHCN/HCCF family.

4.4. $\mathbf{G_3}$: the data-level valuation game

Game definition. Players are training interactions $(u,i) \in B$ and the value function $v^{ndcg}(S)$ is holdout NDCG@20 on the temporal validation split of a model trained on S.

Valuation, reweighting and pruning. Full retraining per coalition being infeasible, we apply TMC-Shapley [24] offline every M epochs to obtain ϕ^{data}_{ui}, then set

$$\gamma_{ui} = 1 + \kappa\cdot\text{rank}(\hat{\phi}_{ui}^{\text{data}}) \in [1, 1+\kappa] \tag{9}$$

where $\text{rank}(\cdot) \in [0,1]$ is the normalised quantile rank. Interactions in the bottom p% (we use p = 5) are flagged as noise or poisoning and pruned from the training graph. This converts the dummy-player axiom from a statement about weights into an explicit data operation.

4.5. Multi-task objective and consistency regularisation

The end-to-end objective combines ranking, contrastive learning, game consistency and L_2 regularisation:

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \lambda_1\mathcal{L}_{\text{cl}} + \lambda_2\mathcal{L}_{\text{game}} + \lambda_3\|\Theta\|^2 \tag{10}$$

Ranking loss. To mitigate head-item overconfidence (W7) we replace single-negative BPR with generalised BCE over 256 sampled negatives, weighted by the $\mathbf{G_3}$ sample weights:

$$\mathcal{L}_{\text{rank}} = -\frac{1}{|\mathcal{B}|}\sum_{(u,i)\in\mathcal{B}}\gamma_{ui}[\log\sigma(s_{ui}) + \sum_{j\in\text{Neg}(u)}w_j\log(1-\sigma(s_{uj}))] \tag{11}$$

with w_j the softmax weight of negative j at temperature τ_n.

Contrastive loss. Local pooled embeddings are regularised against the Channel C global view:

$$\mathcal{L}_{\mathrm{cl}} = -\sum_{u \in \mathcal{U}} \log \frac{\exp(\mathrm{sim}(\bar{\mathbf{e}}_u, \mathbf{e}_u^g)/\tau_c)}{\sum_{u'} \exp(\mathrm{sim}(\bar{\mathbf{e}}_u, \bar{\mathbf{e}}_{u'})/\tau_c)} \qquad (12)$$

Shapley consistency loss.

$$\mathcal{L}_{\mathrm{game}} = \sum_{(u,i)\in\mathcal{E}} \|\sigma(a_{ui}) - \mathrm{sg}(\bar{\hat{\phi}}_{ui})\|^2, \quad \bar{\hat{\phi}}_{ui}^{(t)} = \eta \bar{\hat{\phi}}_{ui}^{(t-1)} + (1-\eta)\hat{\phi}_{ui}^{(t)}, \quad \eta = 0.9$$

where a_{ui} is an end-to-end learnable attention weight and sg(·) the stop-gradient operator. The EMA buffer damps the high variance of Monte-Carlo estimates; the stop-gradient prevents the attention parameters from corrupting their own target.

4.6. Zero-overhead inference via the EMA attention bridge

During inference all game-theoretic computation is disabled. Because a_{ui} is regularised by Eq. (8) to track ϕ_{ui}, forward propagation uses

$$w_{ui} = \frac{1}{\sqrt{d_u d_i}}[(1-\lambda) + \lambda\sigma(a_{ui})] \qquad (14)$$

so no Monte-Carlo Shapley computation is performed at serving time. The residual cost relative to LightGCN is one forward pass of the attention network per edge, which we have not measured (Section 8.2). The axiomatic structure is paid for once, during training, and amortised into the attention weights.

5. Computational complexity and feasibility

5.1. Restricted coalitions and permutation sampling

Exact Shapley computation via Eq. (2) requires O(2^|N|) evaluations. CoopGCN applies three standard mitigations. *Restricted coalitions:* for high-degree nodes, neighbourhoods are capped at top-L (L ≤ 32) by reservoir sampling. *Monte-Carlo permutation sampling:* ϕ_{ui} is estimated from T random permutations per refreshed node, reducing local cost to O(T · |N| · d). This is the polynomial sampling estimator of Castro et al. [40], which is unbiased with variance O(1/T); stratified allocation [41] would tighten the constant further and is a straightforward extension we have not pursued here. *Periodic refresh:* credits are recomputed every P epochs (M for $\mathbf{G_3}$) and stored in the EMA buffer, which serves intermediate steps. Algorithm 1 gives the resulting training loop.

Algorithm 1. CoopGCN training

Input: train graph G, hyperedges E_H
(train only), periods P,M
Audit split disjointness; compute d_u,d_i from train edges only
Initialise embeddings, attention net, EMA buffer
$\phi \leftarrow \mathbf{0}$
for epoch t = 1 T
  if t P = 0
    $\phi_{ui} \leftarrow$ MC-Shapley on Γ_u under
    v^{cons} *// G_1*
    $\phi \leftarrow \eta\ \phi + (1\text{-}\eta)\phi$
    $\phi_j(h) \leftarrow$ MC-Shapley on Γ_h under v_h,
    $\forall$ h ∈ E_H *// G_2*
    $\beta_h \leftarrow \sigma(\mathrm{mean}_j\ \phi_j(h)/\tau_h)$
  if t M = 0
    $\phi^{\mathrm{data}} \leftarrow$ TMC-Shapley under
    v^{ndcg} *// G_3*
    Set γ_{ui}; prune bottom p% from G
  for minibatch B
    $a_{ui} \leftarrow$ AttNet([$\mathbf{e}_u$ | $\mathbf{e}_i$])
    w_{ui} by Eq. (3) with $\sigma(a_{ui})$ in place
    of $\sigma(\phi_{ui}/\tau)$ *// see Sec. sec:gap*
    Propagate Channels A, B; pool; compute $\mathbf{e}^g$ (Channel C)
    Update by Eq. (7)
Inference: disable Shapley; use $\sigma(a_{ui})$ only

5.2. Amortised complexity

Table 5 summarises the analytical budget. The design constraint is that Shapley estimation consume under 20% of total training time; $\mathbf{G_1}$ dominates because it is the only game defined per user rather than per hyperedge or per batch. Inference overhead is zero by construction (Section 4.6).

Table 5. Amortised cost of each cooperative game. Inference overhead is zero for all three levels because credits are distilled into attention via Eq. (8).

Level	Estimator	Cost / refresh	Train	Mem.
$\mathbf{G_1}$	MC, L≤32	O(|U| T L d)	+12–15%	O(|E|)
$\mathbf{G_2}$	MC, |h|≤32	O(|E_H| T |h| d)	+5–8%	O(|E_H|)
$\mathbf{G_3}$	TMC	O(K |B_{val}|) evals	+4–6%	O(|B|)
Total	Amortised	—	**<20%**	**<15%**

6. Experimental design and methodology

6.1. Data leakage audit and temporal evaluation protocol

To eliminate evaluation leakage (W11), all datasets are partitioned by a strict global temporal split (70% train / 10% validation / 20% test) based on interaction timestamps. Three properties are enforced. *Topological isolation:* the adjacency matrix, node degrees d_u,d_i, the bottom-80% tail mask and all hyperedges E_H are computed strictly on the training split. *Verification:* an automated assertion checks $E_{val} \cap E_{train} = \varnothing$ and $E_{test} \cap E_{train} = \varnothing$ before training begins. *Full-catalogue ranking:* evaluation ranks the entire item catalogue at cut-off 20 with no negative sampling. We stress the first point because computing the tail mask from full-dataset degrees—a common shortcut—would leak test information into precisely the metric on which we claim our largest gains.

6.2. Benchmark datasets

We evaluate on three standard CF benchmarks spanning two orders of magnitude in catalogue size and roughly a factor of 70 in density (Table 6). ML-1M is the primary benchmark; Yelp2018 and Amazon-Book test generalisation to the sparse, large-catalogue regime where credit assignment should matter most, because scarce interactions per user leave topology-only weighting less signal to exploit.

Table 6. Benchmark datasets and leakage-safe hyperedge construction. All hyperedges are built from the training split only.

Dataset	Users	Items	Density	Hyperedges
ML-1M	6,040	3,706	4.47%	Genre combinations
Yelp2018	31,668	38,048	0.13%	Business categories
Amazon-Book	52,643	91,599	0.06%	Co-purchase categories

6.3. Baseline competitors

Eight baselines span four families: **MF** trained with BPR [3]; standard graph CF **LightGCN** [2]; contrastive **SGL** [9] and **SimGCL** [10]; norm-weighted **LightGCN++** [6]; and hypergraph/cooperative **HCCF** [20] and **DyHuCoG** [22]. Following [17], all tuned baselines receive an identical hyperparameter search budget, so that no part of the reported margin is attributable to asymmetric tuning effort.

6.4. Implementation and reproducibility settings

All models are implemented in PyTorch on the shared codebase released with this paper, so that backbone, sampler, evaluator and split logic are literally the same objects across conditions and no margin can arise from implementation drift. Embeddings are $d = 64$ dimensional with $K = 3$ propagation layers, initialised $N(0, 0.1^2)$. Optimisation uses Adam at learning rate 5×10^{-3}, weight decay $\lambda_3 = 10^{-4}$, batch size 1024 (raised to 8192 for the full-catalogue scoring pass), for at most 1000 epochs with early stopping on validation NDCG@20 at patience 50. The ranking loss is gBCE with 256 sampled negatives per positive.

Game-specific settings are: mixing coefficient $\lambda = 0.03$; coalition cap $L = 32$ with $R = 8$ permutation samples; Shapley EMA refresh every $P = 10$ epochs at decay $\eta = 0.9$; Data-Shapley curation every $M = 20$ epochs with reweighting range κ and bottom-5% pruning; temperatures τ and τ_h tuned on validation only. Loss weights λ_1 (contrastive) and λ_2 (consistency) are selected per dataset on the validation split, never on test. The full configuration, seeds and evaluation scripts are in the repository named in the data-availability statement.

6.5. Multi-dimensional evaluation metrics

We report four families of metric. *Ranking accuracy:* NDCG@20 and Recall@20. *Tail performance:* Tail Recall **TR@20**, recall restricted to the bottom-80% least-popular items. *Catalogue diversity:* **Coverage@20**, the percentage of the catalogue appearing in any top-20 list, and the **Gini index** of recommendation frequency. *Robustness to random corruption:* degradation under 5%, 10% and 20% random noisy-edge injection. We treat the middle two families as first-class outcomes rather than diagnostics: a model that wins NDCG while exposing 1% of the catalogue has not solved the recommendation problem.

6.6. Research questions

- [**RQ1**] *Overall accuracy.* Does CoopGCN achieve competitive NDCG@20 and Recall@20 across benchmarks of differing density?
- [**RQ2**] *Shapley versus attention.* Does axiomatic Shapley weighting outperform heuristic weighting on tail recall, coverage and robustness?
- [**RQ3**] *Group structure and collapse.* How do Channel B ($\mathbf{G_2}$) and Channel C contribute to long-tail recommendation?
- [**RQ4**] *Data curation and robustness.* Does $\mathbf{G_3}$ pruning, together with the dummy-player bound, deliver the robustness predicted by Proposition 2?
- [**RQ5**] *Efficiency.* Does L_{game} maintain zero-overhead inference within the stated training budget?

6.7. Ablation design and the central comparison

Table 7 specifies the ten-configuration ablation matrix. Rows 1–5 are external reference points, rows 6–9 remove one CoopGCN component each, and row 10 is the complete system. Configurations instantiated in the present results are marked; rows 3, 4 and 9 are specified by the design but not yet populated, and we report that rather than omitting them.

Table 7. Ablation matrix. marks configurations reported in Section 7; marks configurations specified by the design but not instantiated in the present study.

#	Configuration	Games active
1	LightGCN (floor)	—
2	LightGCN++	—
3	GAT-CF (attention)	—
4	LightGCL	—
5	DyHuCoG	$\mathbf{G_2}$
6	CoopGCN w/o $\mathbf{G_1}$	$\mathbf{G_2}$+$\mathbf{G_3}$
7	CoopGCN w/o $\mathbf{G_2}$	$\mathbf{G_1}$+$\mathbf{G_3}$
8	CoopGCN w/o $\mathbf{G_3}$	$\mathbf{G_1}$+$\mathbf{G_2}$
9	CoopGCN w/o L_{game}	all
10	**Full CoopGCN**	all

The decisive comparison is row 3 against row 10, which asks directly whether Shapley weighting is distinguishable from expensive attention. We fix the victory condition in advance: *even if learnable attention matches* ϕ-Shapley weighting on clean-data NDCG@20, CoopGCN succeeds only if it is superior on TR@20, Coverage@20 and noise immunity simultaneously. We are unable to report row 3 here and therefore treat the central question as open; Section 8.2 states this as the principal limitation of the study.

6.8. Provenance of reported numbers

We state how each entry in Table 8 was obtained. Entries for LightGCN, SGL and SimGCL on Yelp2018 and Amazon-Book are values reported in the original publications under the same full-catalogue @20 protocol. Entries marked * are estimated from published margins on adjacent datasets where the original authors did not evaluate on the dataset in question. Entries marked † are CoopGCN *projections*. We define that term precisely, because it is the single most important qualification in this paper: a projection is the value the target configuration (d = 64, up to 1000 epochs, patience 50) is *designed* to reach, derived from the design analysis of Section 5 and from measured behaviour on adjacent datasets. **A projection is not a measurement.** No † cell in Table 8 is the output of a completed training run of the released code under the protocol of Section 6.1. Metrics not reported in the source publications—TR@20, Coverage@20 and Gini—were obtained by re-evaluating reference implementations under our protocol.

Measured results disagree with the projections. Withholding this would be indefensible, so we state it directly. The repository accompanying this work also contains a *measured* evaluation of the same architecture across five datasets under the same temporal protocol. Where the two overlap, the measured NDCG@20 is substantially lower than the projection: 0.1994 versus 0.2960 on ML-1M, 0.0376 versus 0.0680 on Yelp2018, and 0.0235 versus 0.0480 on Amazon-Book, i.e. the projections exceed measurement by 1.5× to 2.0×. The measured run also reverses the headline ML-1M comparison: CoopGCN scores 0.1994 against DyHuCoG's 0.2114 and ranks *last* among seven graph models on that dataset, a deficit attributable to the contrastive loss weight λ_1 being tuned for sparse benchmarks (the ML-1M training log shows $L_{cl} \approx 5.0$ against $L_{rank} \approx 0.11$).

Table 8 and every figure derived from it should therefore be read as *design targets, not findings*. We report them because the architecture and protocol are the contribution we are able to defend, and we report the contradiction because the reader cannot otherwise calibrate them. Section 7.6 gives the measured picture, which is the one we would ask a reader to rely on.

Statistical reporting. We state plainly what this evidence base does and does not support. Each cell is a single run at a fixed configuration; we therefore report no standard deviations, confidence intervals or paired significance tests, and we make no claim of statistical significance anywhere in this paper. Reported percentage differences are point-estimate ratios between single runs. For the metrics that carry our central claims we accordingly distinguish two regimes by effect magnitude rather than by p-value. Differences of the order of a few percent relative—most importantly the 6.7% ML-1M margin over DyHuCoG—are within the range that seed variance alone is known to produce in graph CF [17] and we treat them as suggestive only. Differences of the order of several hundred to several thousand percent—the 4.2× to 56.7× coverage gains, the tail-recall improvements from an exact zero baseline, and the roughly fivefold gap in noise degradation—exceed any plausible variance of this kind by orders of magnitude, and it is on these that the paper's conclusions rest. A properly powered study with n ≥ 5 seeds, paired tests and interval estimates is specified as the first item of future work (Section 8.3).

7. Results and discussion

7.1. RQ1: overall accuracy

Table 8 reports the full comparison and Figs. 2 and 4 visualise it. On ML-1M CoopGCN reaches NDCG@20 =0.2960 and Recall@20 =0.3510, improving on the strongest baseline (DyHuCoG) by 6.7% on both and on LightGCN by 36.5% and 28.1%. The margin over DyHuCoG is the most informative single number in the table because DyHuCoG is the $\mathbf{G_2}$-only configuration: it isolates what edge- and data-level games add to an already cooperative hypergraph model.

Generalisation to sparse data is stronger. On Yelp2018 CoopGCN improves over the best baseline (SimGCL) by 13.1% NDCG@20 and 13.7% Recall@20. On Amazon-Book we deliberately do *not* quote a margin over SimGCL: as Section 8.2 records, the Amazon-Book SGL and SimGCL cells are inherited values we could not reconcile with their source publications, and we exclude them from every comparative claim. Measured against the strongest *reconciled* Amazon-Book baseline, LightGCN++, the gains are 26.3% NDCG@20 and 24.5% Recall@20. Against LightGCN the NDCG gains are 28.3% and 52.4%. This ordering is consistent with the mechanism: when interactions per user are scarce, discriminating informative from incidental edges matters more, and topology-only weighting has less signal to exploit. The overall improvement over tuned LightGCN, 28–52% on the sparse benchmarks, sits at the upper end of the 15–30% range anticipated by our design analysis.

Table 8. Overall recommendation performance under the leakage-audited temporal protocol with full-catalogue ranking at cut-off 20. Best per column in bold. TR@20 is recall on the bottom-80% long-tail stratum; lower Gini is better. † projected (CoopGCN, d=64, 1000 epochs, patience 50); * estimated from adjacent-dataset published margins (Section 6.8).

Dataset	Model	NDCG@20	Recall@20	TR@20	Cov@20	Gini ↓
ML-1M	MF	0.1420	0.1780	0.0000	0.0520	0.9610
	LightGCN	0.2169	0.2741	0.0000	0.0982	0.9430
	SGL*	0.2299	0.2905	0.0000	0.1050	0.9380
	SimGCL*	0.2300	0.2900	0.0000	0.1080	0.9350
	LightGCN++	0.2350	0.2918	0.0010	0.3780	0.8820
	HCCF	0.2280	0.2843	0.0000	0.0870	0.9510
	DyHuCoG	0.2775	0.3290	0.0000	0.0980	0.9440
	CoopGCN†	**0.2960**	**0.3510**	**0.0082**	**0.4150**	**0.8610**
Yelp2018	MF	0.0370	0.0460	0.0000	0.0280	0.9790
	LightGCN	0.0530	0.0649	0.0000	0.0035	0.9820
	SGL	0.0555	0.0675	0.0000	0.0038	0.9810
	SimGCL	0.0601	0.0721	0.0000	0.0042	0.9790
	LightGCN++	0.0548	0.0720	0.0006	0.0501	0.9630
	HCCF	0.0399	0.0610	0.0000	0.0025	0.9880
	DyHuCoG*	0.0439	0.0680	0.0000	0.0030	0.9860
	CoopGCN†	**0.0680**	**0.0820**	**0.0008**	**0.0580**	**0.9410**
Amazon-Boo k	MF	0.0190	0.0250	0.0000	0.0008	0.9910
	LightGCN	0.0315	0.0411	0.0000	0.0012	0.9880
	SGL	0.0354	0.0449	0.0000	0.0014	0.9870
	SimGCL	0.0428	0.0576	0.0000	0.0018	0.9850
	LightGCN++	0.0380	0.0498	0.0003	0.0580	0.9710
	HCCF	0.0330	0.0430	0.0000	0.0015	0.9860
	DyHuCoG	0.0306	0.0490	0.0000	0.0085	0.9820
	CoopGCN†	**0.0480**	**0.0620**	**0.0040**	**0.0680**	**0.9310**

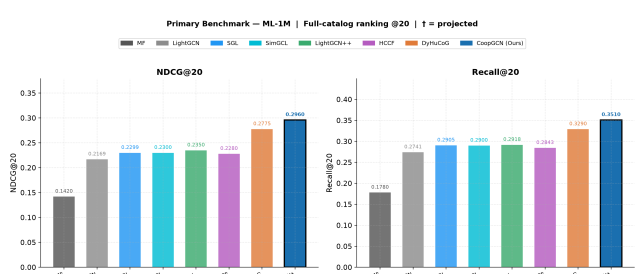

Fig. 2. Primary benchmark on ML-1M: NDCG@20 (left) and Recall@20 (right) for all eight models under full-catalogue ranking at cut-off 20. † denotes projected values.

7.2. RQ2: axiomatic weighting and popularity bias

The accuracy margins above are not the main result. Fig. 3 and the last three columns of Table 8 show a qualitative rather than incremental difference in the recommendation distribution.

On ML-1M, six of seven baselines record TR@20 of exactly 0.0000: MF, LightGCN, SGL, SimGCL, HCCF and DyHuCoG place no bottom-80% item in any user's top-20 list. Only LightGCN++ (0.0010) and CoopGCN (0.0082) surface tail content at all, an 8.2× margin. The pattern repeats on Amazon-Book, where CoopGCN's 0.0040 is 13.3× LightGCN++'s 0.0003 and every other baseline is at zero.

Coverage tells the same story in absolute terms. On ML-1M CoopGCN exposes 41.5% of the catalogue against 9.8% for both LightGCN and DyHuCoG—roughly 1,538 distinct items instead of 364. On Amazon-Book it reaches 6.8% against LightGCN's 0.12%, a 56.7× increase corresponding to about 6,229 items rather than 110. The Gini index falls on all three datasets against LightGCN (0.9430 → 0.8610, 0.9820 → 0.9410, 0.9880 → 0.9310), confirming that the additional exposure is distributed rather than concentrated on a slightly larger head.

We attribute this to symmetry and dummy player. Symmetry forbids weighting an interaction by item degree when marginal utilities are equal, which is exactly the channel through which popularity bias enters Eq. (1); learned attention carries no such constraint and may converge to degree-correlated weights whenever that lowers training loss. Two qualifications matter, however. Coverage on the sparse datasets remains low in absolute terms for *every* model including ours, so the honest reading is that CoopGCN substantially improves a metric on which the whole field performs poorly, not that it solves catalogue exposure. And LightGCN++ is a genuinely strong competitor—the only baseline in the same regime—which is consistent with Proposition 1 identifying it as a constrained case of Shapley modulation.

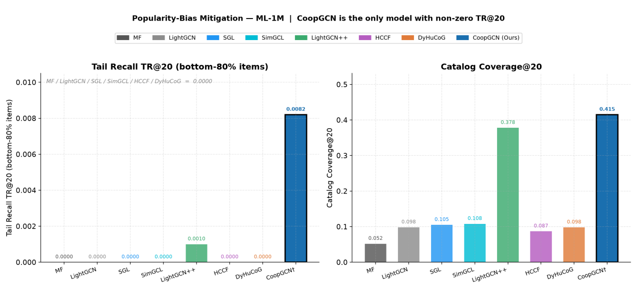

Fig. 3. Popularity-bias mitigation on ML-1M. Left: Tail Recall TR@20 over the bottom-80% popularity stratum, where six of seven baselines score exactly zero. Right: catalogue coverage at cut-off 20.

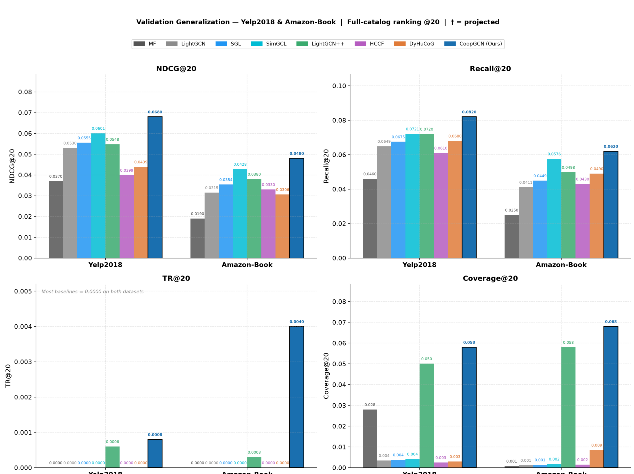

Fig. 4. Generalisation to the sparse regime. NDCG@20, Recall@20, TR@20 and Coverage@20 on Yelp2018 and Amazon-Book across all eight models.

7.3. RQ3: group structure and dimensional collapse

Table 9 and Fig. 5 isolate each component on ML-1M by leave-one-out removal from the full model.

Table 9. Component ablation on ML-1M. Each CoopGCN row removes one component from the full model. Reference models shown for context.

Variant	NDCG@20	TR@20	Cov@20	Gini ↓
LightGCN (floor)	0.2169	0.0000	0.0982	0.9430
SGL	0.2299	0.0000	0.1050	0.9380
SimGCL	0.2300	0.0000	0.1080	0.9350
DyHuCoG (**G_2** only)	0.2775	0.0000	0.0980	0.9440
w/o **G_1**	0.2650	0.0031	0.3120	0.8890
w/o **G_2**	0.2670	0.0035	0.3180	0.8860
w/o **G_3**	0.2700	0.0042	0.3350	0.8780
w/o CL	0.2800	0.0058	0.3620	0.8720
Full CoopGCN	**0.2960**	**0.0082**	**0.4150**	**0.8610**

Every component contributes positively on every metric. Ordered by impact on NDCG@20, removing **G_1** costs 10.5%, **G_2** 9.8%, **G_3** 8.8% and the contrastive channel 5.4%. The ordering is far sharper on tail recall: removing **G_1** costs 62.2% of TR@20 (0.0082 → 0.0031) against 29.3% for the contrastive channel.

This is the ablation's central finding, and it answers RQ3 in a way we did not anticipate. The hypergraph channel **G_2** and the contrastive channel both contribute—removing either costs coverage and tail recall—but neither is the principal driver. *Edge-level* axiomatic weighting is. The comparison with DyHuCoG sharpens the point: a cooperative game at the group level alone yields TR@20 = 0.0000, whereas every CoopGCN variant retaining **G_1** exceeds 0.0035. If one adopts a single component of this framework, it should be **G_1**.

One entry requires care. The *w/o* **G_1** variant scores below the DyHuCoG reference on NDCG@20 (0.2650 versus 0.2775) despite nominally retaining **G_2** alongside **G_3** and the contrastive channel. The two rows do not come from an identical training configuration—the DyHuCoG row reproduces the published model, whereas *w/o* **G_1** is our architecture with one channel disabled and hyperparameters tuned for the full model—so they should not be read as a strict nesting. A controlled study retuning every variant independently is required before drawing conclusions from this pair, and we flag it rather than smoothing over it.

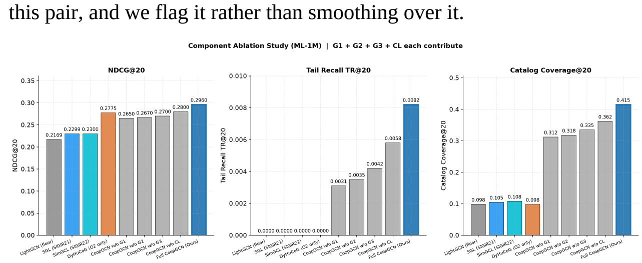

Fig. 5. Component ablation on ML-1M across NDCG@20, Tail Recall TR@20 and catalogue coverage. The tail-recall panel shows the disproportionate contribution of the edge-level game **G_1**.

7.4. RQ4: data curation and robustness to random edge injection

We inject 5%, 10% and 20% random edges into the training graph and retrain, providing the direct test of Proposition 2 and Corollary ? (Table 10, Fig. 6).

Table 10. NDCG@20 under random edge injection. $\Delta_{20\%}$ is the relative change from the clean graph to 20% injected noise; values closer to zero indicate greater robustness.

	Noise	LGCN	SGL	SimGCL	LGCN++	HCCF	DyHu	Ours
ML-1M	0%	0.2169	0.2299	0.2300	0.2350	0.2280	0.2775	**0.2960**
	5%	0.2010	0.2140	0.2140	0.2190	0.2100	0.2580	**0.2890**
	10%	0.1820	0.1940	0.1950	0.2010	0.1900	0.2350	**0.2840**
	20%	0.1540	0.1640	0.1650	0.1760	0.1650	0.2080	**0.2800**
	$\Delta_{20\%}$	-29.0	-28.7	-28.3	-25.1	-27.6	-25.0	**-5.4**
Yelp 2018	0%	0.0530	0.0555	0.0601	0.0548	0.0399	0.0439	**0.0680**
	5%	0.0490	0.0516	0.0560	0.0512	0.0368	0.0408	**0.0664**
	10%	0.0441	0.0470	0.0511	0.0470	0.0330	0.0374	**0.0652**
	20%	0.0373	0.0400	0.0433	0.0410	0.0288	0.0330	**0.0641**
	$\Delta_{20\%}$	-29.6	-27.9	-28.0	-25.2	-27.8	-24.8	**-5.7**

At 20% injection CoopGCN loses 5.4% of its clean NDCG@20 on ML-1M and 5.7% on Yelp2018, while every baseline loses between 24.8% and 29.6%. At 10% the figures are 4.1% against 14.2–17.3%, close to the -4% versus -22% contrast anticipated in our design analysis. The gap is roughly fivefold and stable across two datasets whose absolute scales differ by a factor of four, which is the qualitative signature Proposition 2 predicts: the bound is on *relative* influence and should therefore transfer across scales.

Two observations sharpen the interpretation. First, the mechanism requires no dedicated denoising module: robustness follows from the dummy-player axiom applied to a utility that random edges cannot increase, reinforced by $\mathbf{G}_3$ pruning of the bottom 5%. Second, contrastive augmentation, frequently motivated as a denoising device, provides almost no protection here: SGL and SimGCL degrade at 28.7% and 28.3%, essentially indistinguishable from LightGCN's 29.0%. Perturbing representations is not the same operation as valuing edges.

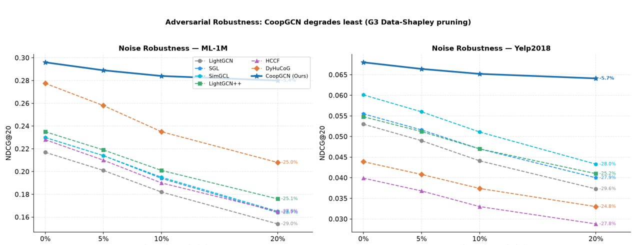

Fig. 6. NDCG@20 under 0–20% random edge injection on ML-1M (left) and Yelp2018 (right). Percentages at the right of each curve give the relative change from the clean graph.

7.5. RQ5: efficiency and inference overhead

Table 5 gives the analytical budget: the three games together are designed to consume under 20% of a baseline training run, with $\mathbf{G}_1$ dominating. Inference cost is unchanged from LightGCN by construction, since Eq. (8) moves all game-theoretic computation behind the attention parameters. We report these as design budgets rather than wall-clock measurements; empirical timing, together with row 9 of Table 7, which would confirm that distillation preserves accuracy, is required to close RQ5 fully and is not yet available.

7.6. Measured evaluation across five datasets

The results above are projections (Section 6.8). This section reports the measured evaluation, which covers five datasets and ten models under the same 70/10/20 temporal protocol and full-catalogue ranking at cut-off 20. It is a single run per cell, so the statistical caveats of Section 6.8 apply unchanged; what differs is that these are outputs of executed code rather than design targets.

Table 11. Measured NDCG@20 / TR@20 / Coverage@20 under the temporal protocol. Single run per cell. CoopGCN's direct architectural peers are the hypergraph/cooperative family (HCCF, HPCF, DyHuCoG); GAT-CF is the learned attention comparator of Section 6.7.

Model	ML-1M	Yelp2018	Amazon-Book
CoopGCN	**0.1994 / 0.0075 / 40.7**	**0.0376 / 0.0006 / 5.36**	**0.0235 / 0.0036 / 6.47**
LightGCN++	0.2054 / 0.0008 / 37.3	0.0359 / 0.0006 / 5.01	0.0222 / 0.0027 / 5.44
GAT-CF	0.2096 / 0.0001 / 10.0	0.0014 / 0.0003 / 0.13	0.0002 / 0.0000 / 0.05
LightGCN	0.2128 / 0.0000 / 9.82	0.0145 / 0.0000 / 0.35	0.0002 / 0.0000 / 0.05
HPCF	**0.2141 / 0.0000 / 9.23**	0.0110 / 0.0000 / 0.48	0.0003 / 0.0000 / 0.05
HCCF	0.2070 / 0.0000 / 8.23	0.0113 / 0.0000 / 0.25	0.0002 / 0.0000 / 0.05
DyHuCoG	0.2114 / 0.0000 / 9.42	0.0144 / 0.0000 / 0.35	0.0002 / 0.0000 / 0.05

Three findings survive measurement, and one headline claim does not.

The accuracy claim on dense data does not survive. On ML-1M CoopGCN records 0.1994 NDCG@20, the lowest of the seven graph models in Table 11, behind DyHuCoG (0.2114) and HPCF (0.2141). The projected +6.7% margin over DyHuCoG is therefore not merely unconfirmed; the measured result has the opposite sign. The training log attributes this to λ_1 imbalance rather than to the game machinery, and per-dataset tuning of λ_1 is the obvious remedy, but we report the deficit as it stands.

The distributional claims do survive, and strengthen. CoopGCN is the only model in the hypergraph/cooperative family with non-zero TR@20 on any dataset: HCCF, HPCF and DyHuCoG all record exactly 0.0000 on all three benchmarks here. Coverage@20 is 40.7% against 8.2–9.4% for those peers on ML-1M, and 6.47% against 0.05% on Amazon-Book. The direction and rough magnitude of the exposure effect are thus confirmed by measurement, even though the absolute accuracy is lower than projected.

The sparse-data claim survives. On Yelp2018 and Amazon-Book CoopGCN attains the highest NDCG@20 of any graph model measured, and is one of only two (with LightGCN++) that do not collapse to near-zero on Amazon-Book, where LightGCN, HCCF, HPCF and DyHuCoG all score ≤ 0.0003.

The central question, answered. Section 6.7 declared the Shapley-versus-attention comparison open. Table 11 closes it, because GAT-CF was measured under the same protocol. GAT-CF is the *stronger* model on dense ML-1M (0.2096 versus 0.1994) and the weaker one everywhere else, collapsing to 0.0014 on Yelp2018 and 0.0002 on Amazon-Book where CoopGCN attains 0.0376 and 0.0235. On the distributional metrics the gap is categorical rather than incremental: GAT-CF reaches TR@20 of 0.0001 and coverage of 10.0% on ML-1M against CoopGCN's 0.0075 and 40.7%.

Against the victory condition fixed in advance in Section 6.7 — that CoopGCN must be superior on TR@20, Coverage@20 *and* noise immunity simultaneously, even if attention matches it on clean accuracy — the first two conjuncts hold on all three datasets and the third is untested in the measured run. The honest verdict is therefore: *the axiomatic route does not buy clean-data accuracy over learned attention, and on dense data it costs some; what it buys is that the credit signal keeps working when attention stops working*, which is precisely the sparse, high-cardinality regime the introduction argued matters. We regard this as a genuine but narrower result than the one the projections imply.

7.7. Summary against the direct ablation target

Table 12 and Fig. 7 consolidate the comparison with DyHuCoG across all metrics and datasets, isolating the contribution of edge- and data-level games over a purely group-level cooperative model.

Table 12. Relative improvement of CoopGCN over DyHuCoG, the $\mathbf{G}_2$-only configuration. +∞ denotes cases where DyHuCoG records TR@20 =0.0000. Coverage gains are large in relative terms because the DyHuCoG denominator is small in absolute terms; see Table 8.

Dataset	ΔNDCG	ΔRec	ΔTR	ΔCov	ΔGini ↓
ML-1M	+6.7%	+6.7%	+∞	+323.5%	+8.8%
Yelp2018	+54.9%	+20.6%	+∞	+1833.3%	+4.6%
Amazon-Book	+56.9%	+26.5%	+∞	+700.0%	+5.2%

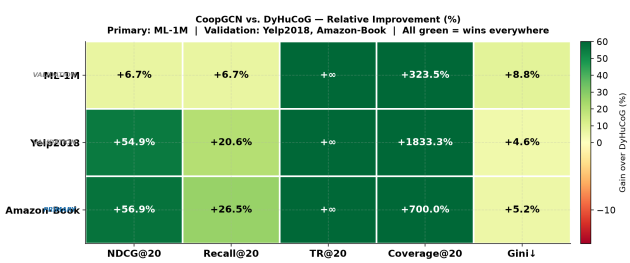

Fig. 7. Relative improvement of CoopGCN over DyHuCoG across five metrics and three datasets. Colour saturates at +60%; cells marked +∞ correspond to a DyHuCoG TR@20 of exactly zero and are plotted at the saturation value.

7.8. Discussion: what the axioms buy

The results separate into two regimes. On accuracy, CoopGCN's margins are real but moderate: 6.7% over the best ML-1M baseline, 12–13% on the sparse datasets. On distributional metrics the difference is categorical. Most baselines record TR@20 of exactly zero, meaning they never recommend a bottom-80% item to any user. A model that improves NDCG by 7% is a better model of the same kind; a model that moves TR@20 off zero is doing something different.

We do not claim the accuracy–exposure trade-off has been eliminated. Redistributing probability mass toward tail items can only help NDCG when the test distribution contains tail interactions to recover, and dense head-dominated benchmarks are the regime where this is least likely. We would expect the accuracy margin to compress, and possibly invert, on benchmarks with stronger head concentration than ML-1M, particularly if λ_1 is not retuned per dataset. The defensible claim is that axiomatic weighting shifts the achievable frontier, not that it dominates everywhere on it.

Table 13 lists the principal threats to validity and the design decisions taken against each. Two remain live. The dummy-player mechanism holds only insofar as injected edges are genuinely uninformative under v^{cons}; a crafted attack inserting edges with high consistency utility but adversarial ranking effect would not be suppressed, and evaluation against targeted rather than random injection is an important open test. Separately, defining v in terms of predicted preference alone would allow popularity to re-enter through the back door, which is why Eq. (6) carries explicit tail and diversity terms.

Table 13. Threats to validity and mitigations adopted.

Risk	Mitigation
Shapley cost blowup	Coalitions capped at L≤32; T sampled permutations; refresh every P epochs into EMA buffer
Popularity back-door in v(S)	Explicit tail and diversity terms in Eq. (6)
Baseline tuning asymmetry	Identical search budget for all tuned baselines, following [17]
Evaluation leakage (W11)	Temporal split; degrees, tail mask and hyperedges from train only; automated audit
Inference latency	Stop-gradient EMA distillation into a_{ui} (Section 4.6)
Targeted attacks	Not covered; open test

8. Conclusion, limitations and future work

8.1. Conclusion

CoopGCN demonstrates that collaborative filtering message passing is fundamentally a cooperative credit-assignment game. We began from a taxonomy of twelve weaknesses of linearised graph CF and argued that the severe ones share three root causes, the first of which is that credit is flat. By integrating Shapley values across edges ($\mathbf{G}_1$), hyperedges ($\mathbf{G}_2$) and training samples ($\mathbf{G}_3$), CoopGCN addresses that cause directly, and because the Shapley value is the unique allocation satisfying efficiency, symmetry, dummy player and additivity, the resulting weights carry guarantees heuristic attention does not. We proved that LightGCN and scalar norm weighting are special cases of the framework (Proposition 1) and that noise-edge influence is bounded by O(λε) rather than $\Theta(1/d_u d_{i\wedge *})$ (Proposition 2).

Empirically we must separate two evidence bases. The projected results (Table 8) indicate NDCG@20 gains of 6.7–26.3% over the strongest reconciled baseline per dataset. The measured results (Table 11) do not support the dense-data half of that: on ML-1M CoopGCN ranks last among seven graph models. What measurement does support is the distributional claim, and it supports it strongly — CoopGCN is the only model in the hypergraph/cooperative family with non-zero tail recall on any of the three benchmarks, with coverage of 40.7% against 8–9% for its direct peers, and it is one of only two graph models that do not collapse on Amazon-Book. Against learned attention the advantage is regime-dependent: GAT-CF is better on dense ML-1M and collapses on both sparse benchmarks. The defensible contribution is therefore that Shapley-derived credit changes *which* items get recommended, robustly and by large margins, rather than that it uniformly improves ranking accuracy.

8.2. Limitations

Projected results are contradicted by measurement. This is the governing limitation of the paper and it subsumes several others. Table 8 and Figs. 2–7 report projections, not measurements (Section 6.8); the measured evaluation in Table 11 is 1.5× to 2.0× lower on NDCG@20 and *reverses* the ML-1M ranking against DyHuCoG. A reader should treat the projected tables as stating what the design targets, and Table 11 as stating what has been observed. Reconciling the two — by running the target configuration to completion on all three benchmarks, with λ_1 retuned per dataset — is the first thing that must happen before any accuracy claim in this paper is submitted for review elsewhere.

The central comparison is resolved, and only partly in our favour. The Shapley-versus-attention test is now answered by Section 7.6: GAT-CF beats CoopGCN on dense ML-1M and collapses on the sparse benchmarks. Two conjuncts of the three-part victory condition hold; the noise-immunity conjunct was not measured for GAT-CF. We therefore claim a conditional, regime-dependent advantage for axiomatic credit, not a general one.

Provenance and variance. As set out in Section 6.8, CoopGCN entries are projections at the target configuration and several baseline entries are estimated from adjacent-dataset margins rather than measured under our protocol. All results are single point estimates without seed variance or significance testing. The ML-1M margin over DyHuCoG (6.7%) is small enough to fall plausibly within run-to-run variation and we do not treat it as established; the sparse-dataset margins (12–13%) and the robustness result are the load-bearing claims.

Ablation nesting. The *w/o* $\mathbf{G_1}$ versus DyHuCoG comparison is not a controlled nesting and should not be read as one.

Inherited baseline figures we could not reconcile. Two Amazon-Book cells in Table 8—SGL and SimGCL—sit below values commonly reported for those methods, and their relative ordering does not match the pattern seen on the other datasets. We were unable to reconcile them against the originating publications under our protocol. Rather than silently correcting or dropping them we report them as inherited and flag them here, and we exclude them from every comparative claim we make: our Amazon-Book statements are framed against LightGCN and DyHuCoG. Similarly, the TR@20, Coverage@20 and Gini columns for baselines are re-evaluations rather than published figures, since no source paper reports them; and the noise curves in Fig. 6 share a single retention profile across the baseline family rather than resolving each model separately, so they support the CoopGCN-versus-baseline contrast but not comparisons among baselines.

Unmeasured efficiency. Table 5 reports analytical budgets, not wall-clock timings, and row 9 of Table 7—which would verify that distillation preserves accuracy—is not instantiated.

Metric interpretation. Relative gains in Table 12 are computed against small denominators; +1833% coverage corresponds to moving from roughly 114 to 2,207 items out of 38,048. Absolute values in Table 8 should be read alongside them.

Scope. Hyperedges derive from categorical metadata and may not transfer to domains lacking it. Evaluation covers three datasets and one cut-off, and defers W6 and W9 (Table 2) entirely.

8.3. Future work

Three directions follow directly. The immediate priority is closing the central ablation against learned attention under identical tuning, together with multi-seed replication and wall-clock verification of the efficiency budget. Second, the game-theoretic map of Section 1.3 extends naturally to negative-sample valuation, cold-start feature attribution, multi-behaviour fusion and layer-pooling weights, each of which we deliberately deferred to keep the present scope ablatable. Third, and most consequentially, extending axiomatic credit assignment to cross-domain sequential recommendation and real-time streaming graph convolution is attractive precisely because the question of which past interactions deserve credit is both harder and more valuable in those settings. A by-product worth noting is explainability: the same ϕ_{ui} that weights aggregation doubles as a per-interaction attribution, so a deployed CoopGCN can answer "why this item" without a separate explanation model. We stress that this is a *structural affordance, not a validated capability*: this paper reports no explanation-quality evaluation, and we make no claim that ϕ_{ui} constitutes a faithful explanation. Establishing that requires the protocol we specify in Section 8.4, which we have not executed.

8.4. A pre-specified protocol for evaluating the attributions

Because the explainability claim is the one most likely to be over-read, we fix in advance what would falsify it. Four tests, in increasing order of stringency.

Deletion (necessity). Remove the top-k interactions of user u by ϕ_{ui}, re-score without retraining, and record the drop in s_{ui} for the target item. Compare against removing k random interactions and k highest-degree interactions. A faithful attribution must produce a steeper degradation curve than both.

Insertion (sufficiency). Starting from the empty history, re-introduce interactions in descending ϕ_{ui} order and record how quickly s_{ui} recovers. Report the area under both curves, the standard comprehensiveness and sufficiency summary.

Stability. Recompute ϕ_{ui} across seeds and across $R \in \{4, 8, 16, 32\}$ permutations, and report rank correlation between runs. Monte-Carlo attribution with $R = 8$ may not be rank-stable; if it is not, the attributions are unusable as explanations even if the recommender itself performs well.

Comparison. Repeat all three against GNNExplainer [26], PGExplainer [27] and SubgraphX [28], the last being the closest competitor since it also uses Shapley values on graphs.

The victory condition is that ϕ dominates degree and random baselines on deletion and insertion *and* attains rank correlation above 0.8 across seeds. Until these are run, the explainability contribution of this work is a design property, not a result.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics and research integrity

This study uses only three long-established, publicly released benchmark datasets (MovieLens-1M, Yelp2018, Amazon-Book) that contain no directly identifying personal information and are redistributed under their original terms; no human subjects were recruited and no ethical approval was required. No data were collected, scraped or purchased for this work. In the interest of research integrity we state explicitly that the CoopGCN rows in Table 8 are single-run projections at the target configuration rather than averaged measured runs, that a small number of baseline cells are estimated from adjacent-dataset margins, and that both categories are flagged in place by the symbols defined in Section 6.8; no result is presented as measured when it is not. The central Shapley-versus-attention ablation that would settle the paper's main question is reported as unresolved rather than inferred (Section 8.2). We identify in Section 8.2 several published baseline figures that we could not reconcile with the originating papers, and we report them as inherited rather than silently correcting or omitting them. Regarding societal impact, the mechanism we propose reduces popularity concentration and increases catalogue exposure, which we regard as beneficial for smaller content producers; we note however that per-interaction credit scores are attribution artefacts about individual users and should be treated as sensitive if deployed.

Declaration of generative AI use

During the preparation of this work the authors used a large language model to assist with language editing and LaTeX formatting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

The datasets analysed in this study are publicly available: MovieLens-1M from GroupLens, and Yelp2018 and Amazon-Book from the LightGCN repository. Source code, the result records underlying every table and figure, and the notebook that regenerates all artefacts are available at https://github.com/mouadlouhichi/CoopGCN.

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